NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-22/13A
10 September 1964
SUPERSEDING
MIL-R-22/13
21 March 1962

PERFORMANCE SPECIFICATION

RESISTOR, VARIABLE, (WIREWOUND, POWER TYPE)

(UNENCLOSED), STYLE RP50

This specification sheet forms a part of the latest issue of Military Specification MIL-R-22

REQUIREMENTS:

Dimension and configuration: See figure and Table I.

Resistance value: See table I.

Resistance tolerance: ±10 percent.

Electrical off position: All three off positions applicable. See table II.

Shaft style and type of mounting: See table III.

Shaft length: See table IV.

Mechanical rotation: $330^{o} \pm 5^{o}$

Hardware: Two 1/4-20 UNC-2A flathead machine screws of sufficient length to permit mounting of a 0.250-inch thick panel shall be supplied.

Dielectric withstanding voltage:
At atmospheric pressure - Method 301 of MIL-STD-202 test voltage - 1,000 volts rms.

At reduced pressure - Method 105 of MIL-STD-202 test voltage - 550 volts.

DC resistance: Total and minimum. Method 303 of MIL-STD-202.

Total resistance within ±10 percent of the nominal total resistance specified in table I.

Minimum resistance shall not exceed 0.2 percent of the nominal total resistance value or 0.2 ohm, whichever is greater.

Torque:
Operating - Not less than 1 pound-inches (lb-in), nor more than 10 pound-inches (lb-in). Stop - 15 lb-in. Locking - not applicable.

Low temperature exposure: Torque not in excess of 10.5 lb-in.

Power rating: 750 watts at 25°C. Mounted on a steel panel 12 inches square by 0.063 inches thick. Hot spot not in excess of 390°C.

Load life: Full wattage (see power rating above) applied 1-1/2 hours on and 1/2 hour off for 1,000 hours. Change in total resistance not to exceed ± 5 percent.

Vibration: Not applicable.

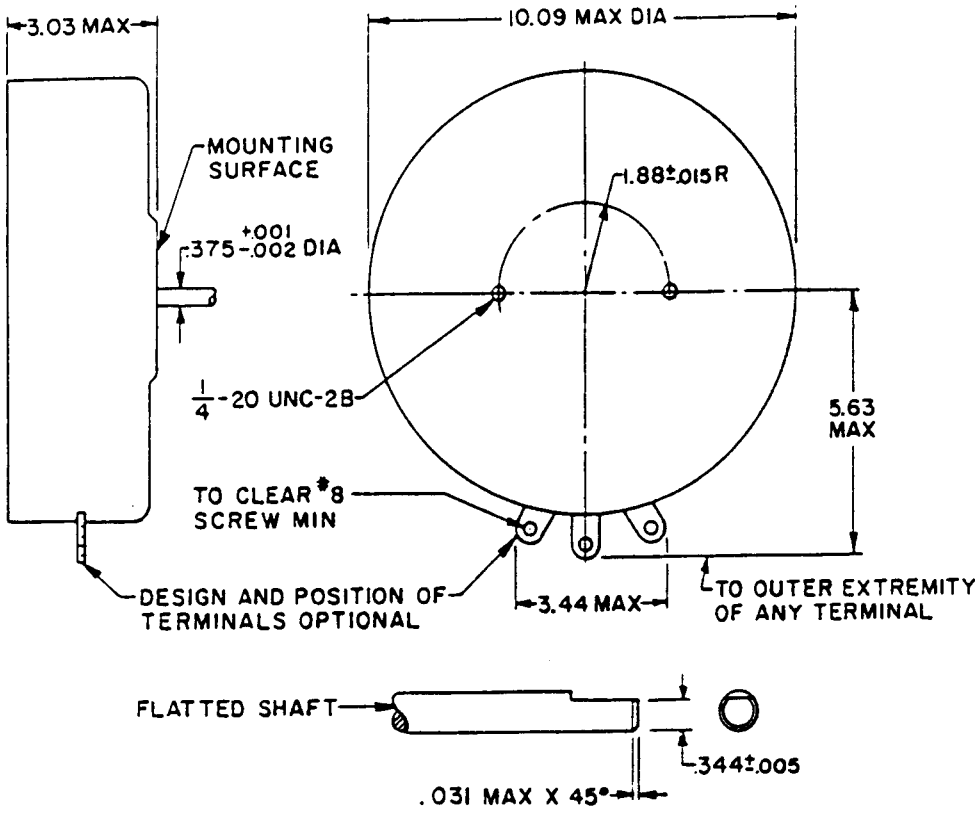
Salt spray (corrosion): Method 101 of MIL-STD-202. Test condition letter B (48 hours).

Humidity (steady state): Method 103 of MIL-STD-202. Test condition letter B (96 hours). Change in total resistance not to exceed ± 10 percent.

Life (rotation): Method 206 of MIL-STD-202.

FSC 5905

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INCHES	MM
.001	.03
.002	.05
.005	.13
.015	.38
.031	.79
.375	9.53
1.88	47.75
3.03	76.96
3.44	87.38
5.63	143.00
10.09	256.29

NOTES:
1. All dimensions in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm to the nearest .01 millimeter.

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Table I— Style RP50.

Type designation 1/ (for reference only)	Nominal total resistance value (ohms)	Maximum current (not to be exceeded on any portion of the winding) (amperes)	Part number 1/
RP50---2R0KK	2.0	19.36	M22/13-0001---
RP50---2R5KK	2.5	17.32	M22/13-0002---
RP50---3R0KK	3.0	15.81	M22/13-0003---
RP50---4R0KK	4.0	13.69	M22/13-0004---
RP50---5R0KK	5.0	12.25	M22/13-0005---
RP50---7R5KK	7.5	10.00	M22/13-0006---
RP50---100KK	10	8.66	M22/13-0007---
RP50---120KK	12	7.90	M22/13-0008---
RP50---150KK	15	7.07	M22/13-0009---
RP50---250KK	25	5.48	M22/13-0010---
RP50---350KK	35	4.63	M22/13-0011---
RP50---500KK	50	3.87	M22/13-0012---
RP50---750KK	75	3.16	M22/13-0013---
RP50---101KK	100	2.74	M22/13-0014---
RP50---151KK	150	2.24	M22/13-0015---
RP50---201KK	200	1.94	M22/13-0016---
RP50---251KK	250	1.73	M22/13-0017---
RP50---351KK	350	1.46	M22/13-0018---
RP50---501KK	500	1.23	M22/13-0019---
RP50---751KK	750	1.00	M22/13-0020---
RP50---102KK	1,000	0.87	M22/13-0021---
RP50---122KK	1,200	0.79	M22/13-0022---
RP50---152KK	1,500	0.71	M22/13-0023---
RP50---202KK	2,000	0.61	M22/13-0024---
RP50---252KK	2,500	0.55	M22/13-0025---

1/ The complete type designation and part number include symbols indicating electrical off position from table II, style of shaft and type of mounting from table III, and length of operating shaft from table IV. (See example of part number.)

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Table II— Electrical off position.

Symbol	Electrical off position
1.....	No electrical off position.
2.....	Electrical off position at end of rotation of control knob in a counterclockwise direction.
3.....	Electrical off position at end of rotation of control knob in a clockwise direction.

Table III— Style of shaft and type of mounting.

Symbol	Style of shaft	3/8-inch diameter shaft 2-hole mounting
F.....	Flatted	X

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Table IV— Length of operating shaft.

Standard length of shaft measured from mounting surface of resistor.	
Flatted	
Symbol	3/8-inch diameter shaft
	2-hole mounting
	Inches, ±3/64
A.....	---
B.....	---
D.....	7/8
G.....	---
H.....	1-1/2
J.....	2
K.....	1/ 2-1/2
N.....	4
R.....	6

1/ This shaft shall be flatted to within 0.156 inch of the mounting surface.

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Test condition letter C (5,000 cycles) for two-hole mounting types. Change in total resistance not to exceed ±5 percent.

Terminal strength: Not applicable.

Shock: Not applicable.

Part number: Consists of the basic number of this specification sheet, a dash number from table I and electrical and mechanical characteristics taken from tables II, III, and IV.

Example:

M22/13-	0001	1	F	H
Basic document number (indicating MIL-R-22/13)	Dash number from table I (indicating 2.0 ohms)	Electrical off position from table II (indicating off position not used)	Style of shaft and type of mounting from table III (indicating flatted shaft)	Length of shaft from table IV (indicating 1-1/2 inch shaft)

Custodians:
Army - EL
Navy - SH
Air Force - 11

Reviewers:
Army - MI, MU
Navy - SH
Air Force - 11, 17, 85

Users:
Army - MO

Preparing activity:
Navy - SH

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